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Offline ASPA - based BGP export Policy Checker

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Problem Description

Border Gateway Protocol (BGP) enables communication between Autonomous Systems (ASes) across the Internet. Incorrect export policies can unintentionally advertise routes to external neighbours, potentially causing BGP route leaks.

ExportGuard is a proposed offline checking tool that analyses sample BGP routes, export policies, AS_PATH information, and locally available ASPA data. It identifies suspicious route advertisements and generates an explainable report before deployment.

Problem

- Manual inspection of complex BGP export policies is time-consuming.
- Incorrect route propagation can cause routing security issues.
- ASPA validation requires correct interpretation of provider relationships and available data.
- Operators need clear explanations instead of simple pass/fail results.

Background

1. Border Gateway Protocol (BGP)
BGP exchanges routing information between different Autonomous Systems on the Internet.

2. AS_PATH
AS_PATH records the sequence of Autonomous Systems through which a route has travelled.

3. Autonomous System Provider Authorization (ASPA)
ASPA uses signed RPKI objects to identify the provider ASes authorised by a customer AS.

4. BGP Export Policy
An export policy determines which routes a network advertises to its external BGP neighbours.



BGP(Border Gateway protocol)

Exchanges routing information between networks.



AS_PATH

Records the path taken by a route.



ASPA (Autonomous System Provider Authorization)

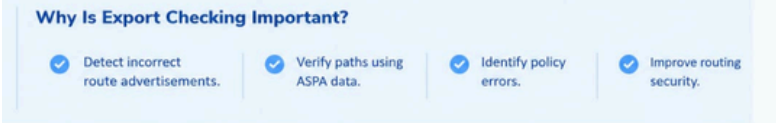
Verifies provider relationships using signed RPKI data.



BGP Export Policy

Controls which routes are advertised to external networks.





Literature & RFC Review

1. **RFC 9234 :** BGP Roles and OTC Supports route-leak prevention and detection.
2. **ASPA-Based AS_PATH Verification:**
Checks AS relationships and suspicious rotes.
3. **BGP Export Verification:**
Identifies misconfigurations before route advertisement.
4**. Identified Opportunity:**
ExportGuard focuses on making these checks easier to review through:

- Explainable rule-based findings.
- Local, offline test scenarios.
- A route-by-route risk report.
- Before-and-after comparison of proposed policy changes.

Proposed Solution

We propose ExportGuard is an offline BGP Export Policy Checker designed to identify potential routing risks before BGP routes are advertised to neighbouring networks. It analyses AS_PATH information, export policy rules, and locally available ASPA data to detect suspicious route advertisements. The tool validates inputs, checks policies, and generates clear, explainable reports highlighting affected routes, detected issues, and supporting evidence. By working offline, ExportGuard helps network operators test and review export policies before deployment without relying on a live router connection.

- Offline policy validation without connecting to a live router.
- ASPA-based path checking using available local data.

Optimization Proposed

ExportGuard automates offline BGP policy checking, identifies routing risks, explains results, and compares policy changes.

Feature	Traditional/Manual Review	ExportGuard(proposed)
Policy Checking	Manual inspection	Repeatable offline validation
Findings	Reviewer-dependent results	Structured, rule-based findings
Explanation	Manual investigation required	Route-specific evidence and reasons
Missing ASPA Data	Further investigation may be needed	“Unknown / Needs Review” status
Policy Comparison	Difficult to compare consistently	Before-and-after report comparison

Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.

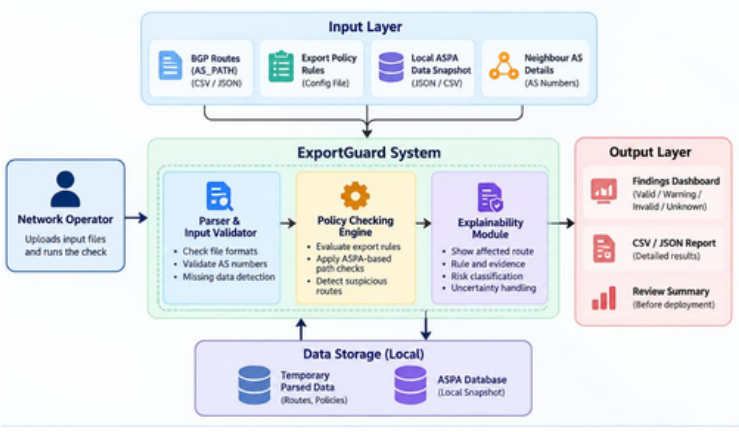


Figure: System Architecture UML Diagram

Data Flow Diagram

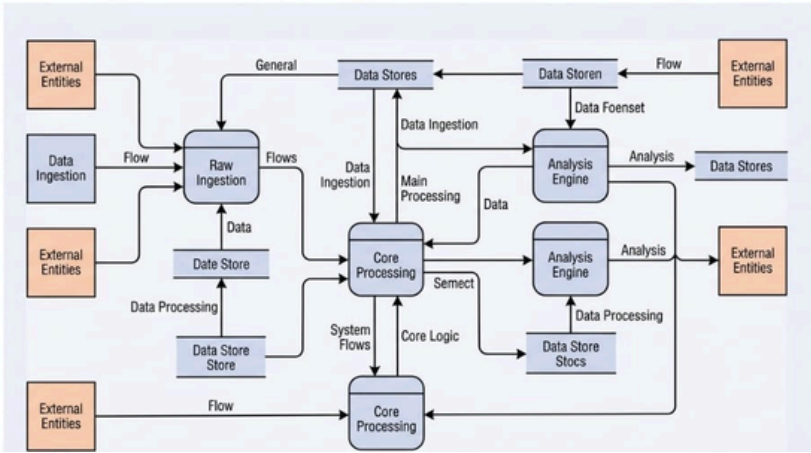


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Testing methodology

- **Functional testing:** Check rules.
- **Edge-case testing**: Test empty files
- **Consistency testing**: Compare outputs.
- **Comparison testing**: Check policy changes.
- **Offline testing**: Test using local files.



Bibliography

[1] **RFC 9234**
Route Leak Prevention and Detection Using Roles in UPDATE and OPEN Messages. IETF, 2022.

[2] **IETF SIDROPS Working Group**
BGP AS_PATH Verification Based on Autonomous System Provider Authorization (ASPA) Objects. Internet-Draft.

[3] **IETF SIDROPS Working Group**
ASPA-Based AS_PATH Verification for BGP Export. Internet-Draft.

[4] **Project Documentation**
ExportGuard architecture, input format, test cases, and measured prototype results.

Thank You for Your Attention!